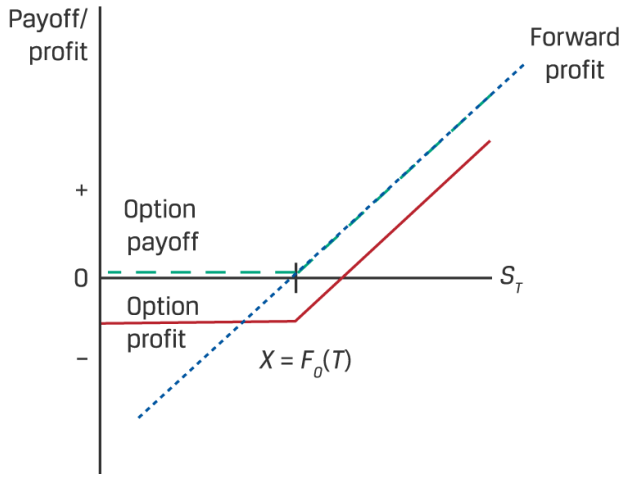
FORWARD COMMITMENTS VS. CONTINGENT CLAIMS

Learning Outcome

- contrast forward commitments with contingent claims

A firm commitment requires both counterparties to perform under a derivative contract, while an option buyer can decide whether to perform under the contract at maturity depending on the underlying price relative to the exercise price. Market participants often create similar exposures to an underlying using these different derivative instrument types. For example, both a long forward position and a long call option position will gain from an increase in the underlying price. Exhibit 12 contrasts the payoff and profit of these two derivative contracts, where the exercise price, X, equals the forward price, $F_0(T)$.

Exhibit 12: Long Forward and Long Call Option Payoff Profiles



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As shown earlier, both the long forward and the call option payoffs increase as S_T rises. In the case of a forward, this linear relationship is equal to $[S_T - F_0(T)]$, with the payoff equal to profit because no cash is exchanged at inception. For the buyer of a call option with an exercise price of $F_0(T)$, Equation 2 changes to

$$\Pi = \max[0, S_T - F_0(T)] - c_0.$$

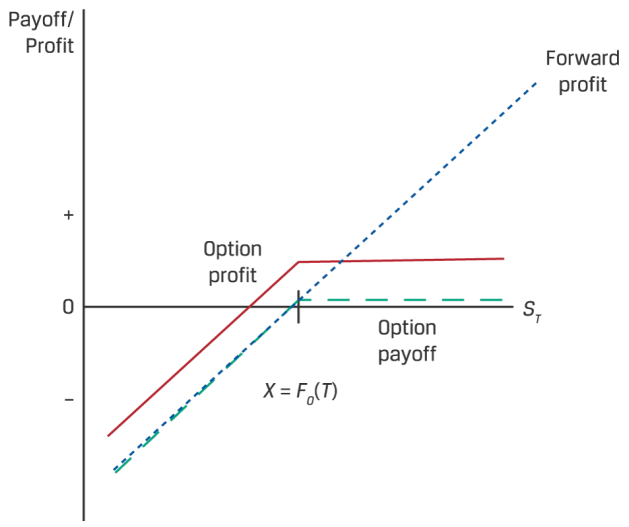
Setting the forward payoff/profit $[S_T - F_0(T)]$ equal to the call option profit, Π, gives us the following relative profit profile between the forward and option:

- $S_T - F_0(T) > -c_0$ Forward profit exceeds option profit
- $S_T - F_0(T) = -c_0$ Forward profit equals call option profit
- $S_T - F_0(T) < -c_0$ Option profit exceeds forward profit

The side-by-side comparison in Exhibit 12 between the forward and call option profit diagrams shows the long call option's similarity to a long position in the underlying with downside protection in exchange for paying a premium.

Another contingent claim that benefits from a rise in the underlying price is the sold put option. While the long call option and long forward payoffs both rise when the underlying price is above the exercise price, the put option seller's profit is limited to the upfront

Exhibit 13: Long Forward and Short Put Option Payoff Profile



▶ Expand / collapse Extended Description

We can compare the long forward payoff/profit of $[S_T - F_0(T)]$ to a modified version of Equation 6:

$$\Pi = -\max[0, F_0(T) - S_T] + p_0.$$

Setting the forward profit $[S_T - F_0(T)]$ equal to the put option profit, Π, gives us the following relative profit profile between the forward and option:

- $S_T - F_0(T) > p_0$ Forward profit exceeds option profit
- $S_T - F_0(T) = p_0$ Forward profit equals option profit)
- $S_T - F_0(T) < p_0$ Option profit exceeds forward profit

The side-by-side comparison in Exhibit 13 of the forward and sold put option profit diagrams shows the sold put option's similarity to a long position in the underlying, with gains from price appreciation forgone in exchange for receiving a premium. The apparent symmetry between long call and short put positions and the long forward position will be examined in greater detail in a later lesson.

Question Set: Firm Commitments and Contingent Claims

1. Determine the correct answers to fill in the blanks: A _____ forward position, a _____ call option position, and a _____ put option position will gain from a decrease in the underlying price.

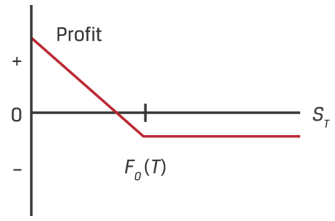
Hide Solution

Solution:

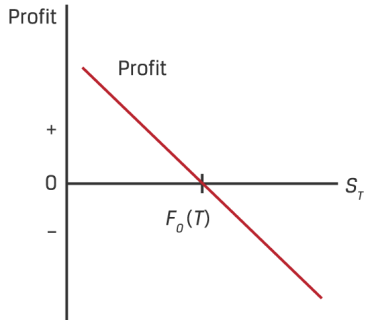
A *short* forward position, a *short* call option position, and a *long* put option position will gain from a decrease in the underlying price.

2. Identify the derivative positions that correspond to the following profit profiles at maturity assuming that the exercise price (X) equals $F_0(T)$:

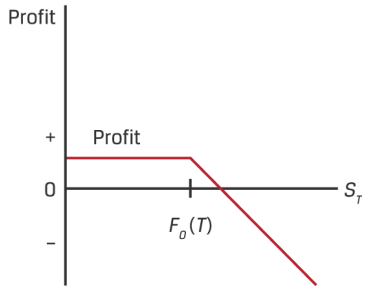
A. Short forward position	1.



B. Long put position 2.



C. Short call position 3.



Hide Solution

Solution:

1. The correct answer is B.

2. The correct answer is A.

3. The correct answer is C.

3. Describe the point at which a short forward and a long put with an exercise price (X) equal to the forward price, $F_0(T)$ have the same profit.

Hide Solution

Solution:

The short forward and long put positions will have the same profit when $F_0(T) - S_T = -p_0$. A short forward position gains from a decline in the underlying price with a payoff/profit of $[F_0(T) - S_T]$, and a long put position has a profit of $\Pi = \max[0, F_0(T) - S_T] - p_0$, from Equation 4, with an exercise price equal to the forward price, $X = F_0(T)$.

OPTIONAL UNTESTED SEGMENT

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The Bloomberg FXFA function allows you to view FX forward rates consistent with the spot rate and interest rate in each currency. This is a typical use case of forward contracts.

> "FXFA <GO> allows you to analyze the pricing relationships between currency spot, forward, and interest rates to identify passive arbitrage opportunities. FXFA helps you determine whether you would get a higher return or lower borrowing cost by executing an FX swap trade instead of lending or borrowing directly in the market." -Bloomberg

To explore this function, type FXFA into the Bloomberg command line and press Enter or <GO>. The "Same Currency" tab allows you to identify passive arbitrage opportunities between two currencies. You can set up the calculation on the "Multi-Currency" tab to display the rates for the currencies and term that are most relevant to your position. The "Central Bank" tab allows you to build new interest rate models for market and implied FX currency swaps so you can test adjustments to your cash flows based on shifting interest rate changes.

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Please note that the first line in the paragraph before Exhibit 13 was supposed to be "Another contingent claim that benefits from a rise in the underlying price is the Short put option." It is *Short*, not Sold.

Created 3 months ago by Obinna Samuel Agwu

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